CS2881 Mini-project

Project Overview

Goal: Practice *reproducing* and *critically examining* results in AI safety research

Project: Choose one paper from the provided options (or propose your own) and recreate the “headline figure”

Project Overview

1. **Reproduce the headline result**
    - Recreate the central figure or table from the paper.
    - Ensure your code and setup are clear, well-documented, and runnable.
2. **Explore divergences**
    - Conduct 1–2 variations on the original experiment.
    - These may involve testing robustness (e.g., shifting datasets, tweaking hyperparameters, or modifying the setup) or applying the method in a new setting.
    - The goal is to probe how fragile or generalizable the original finding is

Project Structure

Team size: 2–4 students.

Deliverables:

Short report (3-5 pages) + github

Recreated headline result (figure or table).

1–2 extensions (divergences).

Reflection on robustness / fragility of findings.

Code + README.

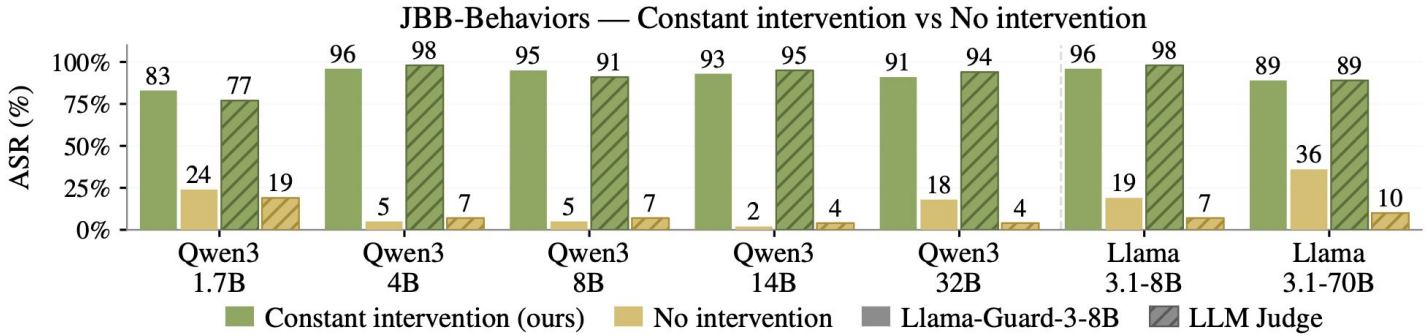
Notes

- If source code is publicly available, divergences/extensions should be non-trivial
- Feel free to propose variations to make things easier in practice
    - Eg, scaling down model size or substituting model series

A Single Neuron Is Sufficient to Bypass Safety Alignment in Large Language Models [link]

Main idea: Finding and patching a single neuron on variety of models leads to big increase in the Attack Success Rate (ASR).

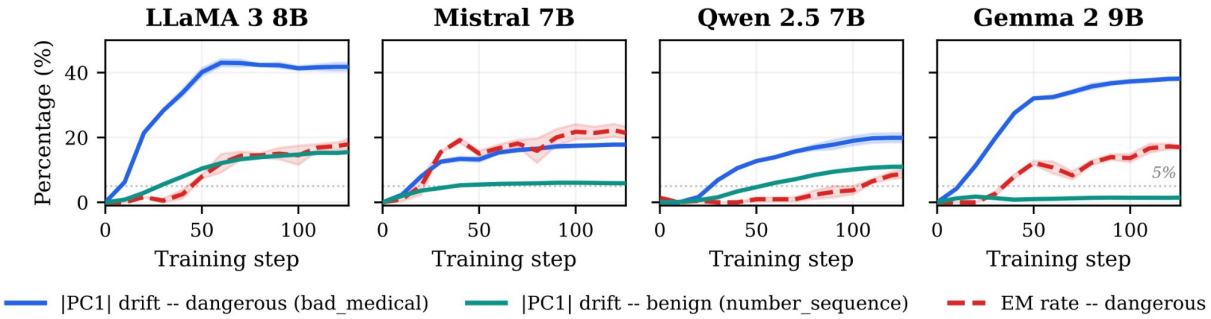
Reproduce: Figure 2 - subset of the models



Trait-space Monitoring for Emergent Misalignment During Supervised Finetuning [link]

Main idea: Monitors activation drift during finetuning to detect emergent misalignment before it shows up behaviorally, using a lightweight per-checkpoint alarm

Reproduce: Figure 1



Retracing the Past: LLMs Emit Training Data When They Get Lost [link]

Main idea: Paper finds that LLMs emit training data just after a spike of uncertainty of the next token. They engineer those spikes more deliberately to increase the success of this type of attack

Reproduce: Table 1

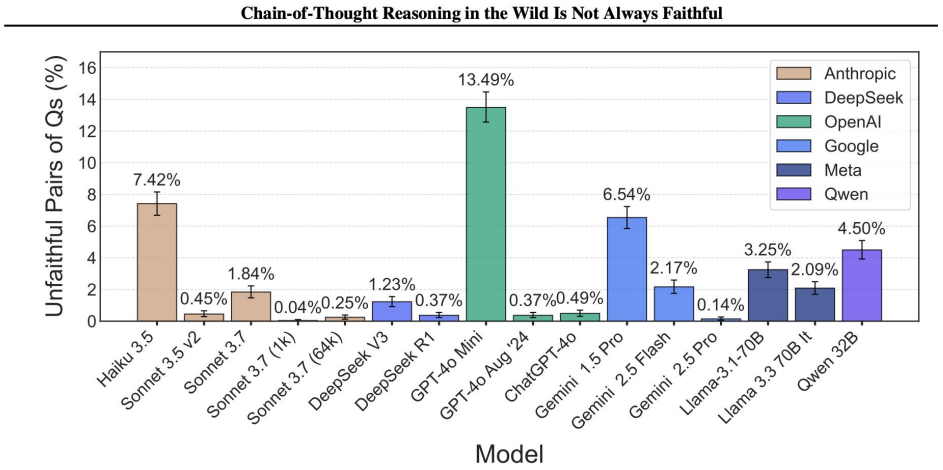
Attack Method	LLAMA 1 (65B)			LLAMA 2 (70B)			OLMO (7B)		
	VM@50	M5@50	M10@50	VM@50	M5@50	M10@50	VM@50	M5@50	M10@50
RA (Nasr et al., 2025)	0.0	0.0	0.0	0.2	0.2	0.2	0.0	0.0	0.4
EA (Nie et al., 2024)	1.0	1.2	1.2	0.8	1.2	1.4	0.2	0.2	0.2
RWA (Nasr et al., 2025)	7.0	8.8	9.6	8.6	10.0	10.6	1.4	2.2	2.2
RSA (Ours, baseline)	7.4	9.4	10.4	10.8	13.2	13.6	1.2	1.6	2.2
CIA (Ours)	**16.0**	**19.0**	**20.0**	**22.2**	**25.4**	**27.0**	**6.0**	**9.4**	**9.6**

Table 1: Attack success rates (%) on unaligned models. We report verbatim matches requiring 50 consecutive tokens (VM@50), and matches allowing up to 5 (M5@50) or 10 (M10@50) token mismatches. The best performing result for each metric and model is highlighted in bold.

Chain-of-Thought Reasoning in the Wild Is Not Always Faithful [link]

Main idea: Evaluates the cases when model gives contradictory answers to the same conceptual question and analyzes the logical errors in the CoT (post hoc rationalization)

Reproduce: Figure 2



Example

Size	Model	ROUGE-L	BLEU	F1	BERTScore
7b	Llama2-Chat-7b	**80.369**$_{\pm 1.679}$	**71.064**$_{\pm 2.033}$	83.415$_{\pm 1.375}$	**94.771**$_{\pm 0.301}$
	Mistral-Instruct-7b	79.121$_{\pm 0.653}$	68.426$_{\pm 0.857}$	**83.741**$_{\pm 0.446}$	94.114$_{\pm 0.134}$
≈13b	SOLAR-10.7b	46.109$_{\pm 3.55}$	38.595$_{\pm 3.677}$	51.224$_{\pm 3.302}$	88.148$_{\pm 0.706}$
	Llama2-Chat-13b	**83.597**$_{\pm 1.104}$	**75.535**$_{\pm 1.404}$	**85.806**$_{\pm 0.882}$	95.184$_{\pm 0.216}$
	Vicuna-13b	70.457$_{\pm 2.444}$	63.59$_{\pm 2.804}$	74.141$_{\pm 2.241}$	93.801$_{\pm 0.507}$
	Mixtral-Instruct-8x7b	80.862$_{\pm 1.226}$	70.697$_{\pm 1.501}$	85.725$_{\pm 0.979}$	**95.686**$_{\pm 0.232}$
	WizardLM-13b	74.923$_{\pm 2.399}$	66.468$_{\pm 2.468}$	77.355$_{\pm 2.279}$	92.759$_{\pm 0.517}$
≈70b	Llama2-Chat-70b	89.567$_{\pm 0.958}$	83.374$_{\pm 1.308}$	90.416$_{\pm 0.772}$	96.436$_{\pm 0.174}$
	Qwen1.5-Chat-72b	**99.154**$_{\pm 0.348}$	**98.412**$_{\pm 0.54}$	**99.138**$_{\pm 0.286}$	**99.757**$_{\pm 0.072}$
	Platypus2-Instruct-70b	83.383$_{\pm 2.235}$	80.693$_{\pm 2.39}$	83.884$_{\pm 2.125}$	96.15$_{\pm 0.463}$

Table 1: We scalably test the vulnerability of instruction-tuned LMs of different sizes against our attack. LMs with higher text similarity scores are more prone to output retrieved text verbatim. We show that LMs with stronger abilities are more vulnerable to prompt-injected data extraction: As model size increases, the maximum values for each size under each metric also increase. Notably, Llama2-Chat-7b can reach a ROUGE score over 80 and a BLEU

Model	ROUGE-L	BLEU	BERTScore	Token F1
7B Models				
Llama-3.1-8B	90.259$_{\pm 5.422}$	88.658$_{\pm 5.766}$	96.207$_{\pm 1.402}$	91.181$_{\pm 4.516}$
Mistral-7B	**48.505**$_{\pm 6.968}$	38.256$_{\pm 6.992}$	**89.178**$_{\pm 1.399}$	55.116$_{\pm 6.781}$
Qwen2.5-7B	77.503$_{\pm 6.160}$	71.433$_{\pm 7.011}$	94.904$_{\pm 1.165}$	82.353$_{\pm 4.756}$
13B Models				
Mixtral-8x7B	**65.899**$_{\pm 4.343}$	51.857$_{\pm 4.474}$	**92.720**$_{\pm 0.763}$	72.904$_{\pm 3.223}$

Table 1: Reproduced Table 1 from Qi et al. (2024) showing RAG training data leakage for 7B and 13B model sizes.

Extensions: cross lingual extraction, summarization as prevention mechanism and varying the RAG pipeline

Timeline

Due date: 10/18

Paper choice + group selection: 10/04; add yourselves to a spreadsheet that will be sent on Slack. You can work with people from other sections.

TFs will host OH/checkins

TLDR

- We don’t care about exact reproduction as much as getting at the main idea
- Feel free to propose variations and/or interesting extensions!